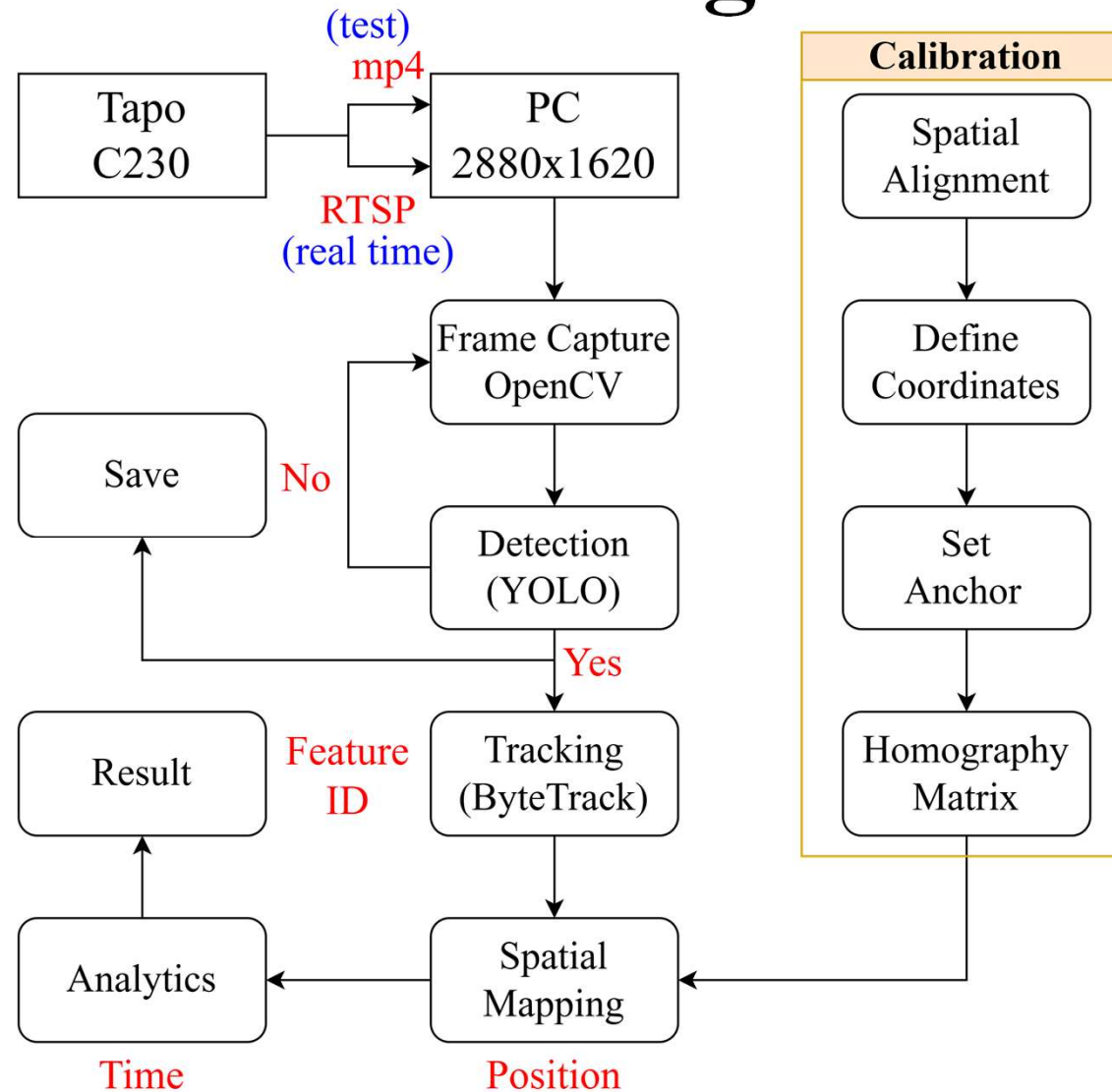


Calibration

日期:2026/7/24 報告人:陳眾科

System Architecture Diagram



Calibration-(Spatial Alignment)

<https://github.com/ChenChungKo/person-detection-and-tracking>

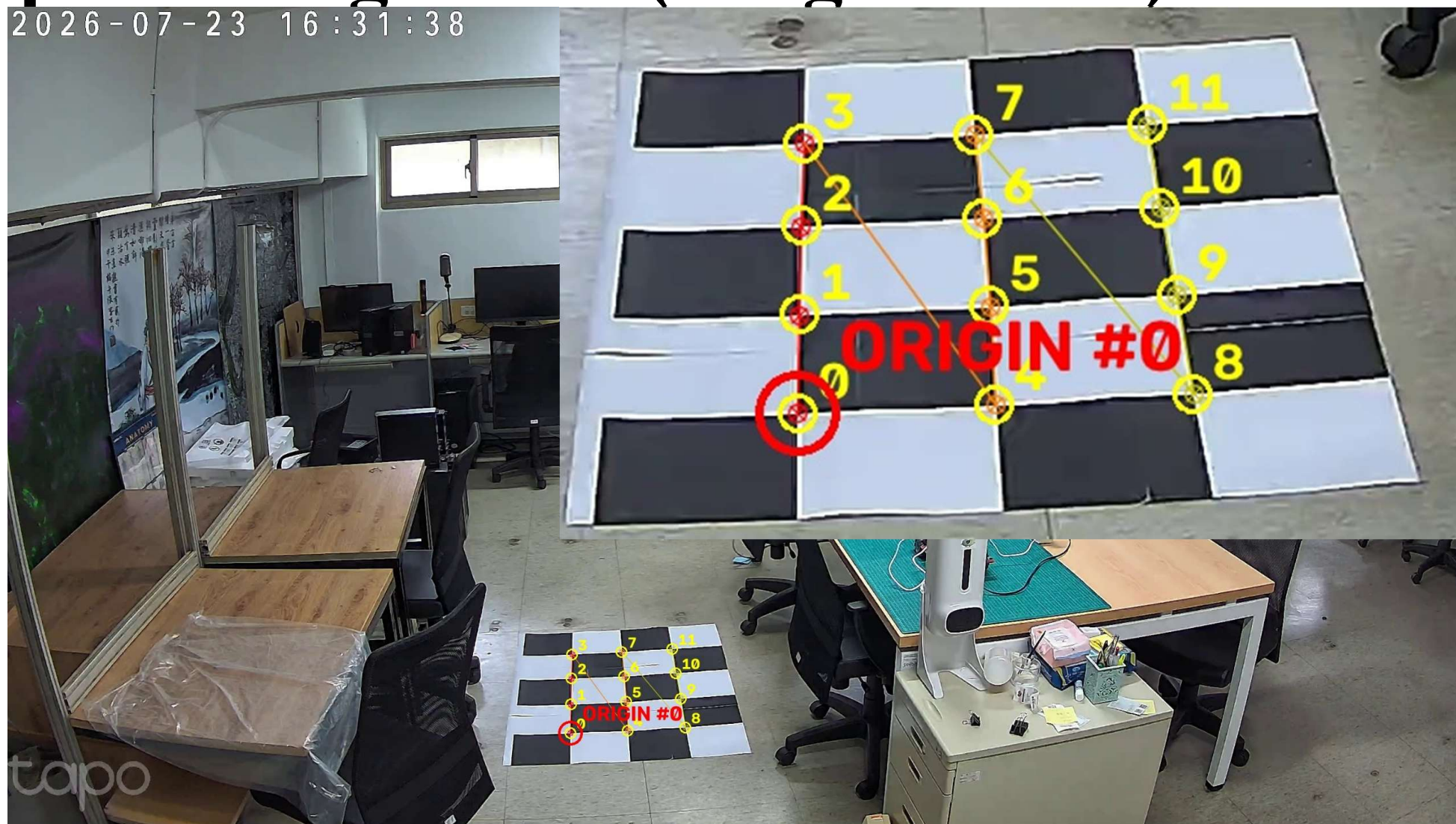
版本	檔案	作法	備註
v1	<code>calibration/homography_v1_manual.json</code>	手動點選磁磚角 (<code>calibrate_boundary.py</code>)	點擊量測誤 差約 8.9 cm
v2	<code>calibration/homography_v2_chessboard.json</code>	地板大棋盤格自動角點 (<code>calibrate_chessboard_floor.py</code>)	目前預設； 點擊量測誤 差約 3.8 cm

Spatial Alignment-(Manual)

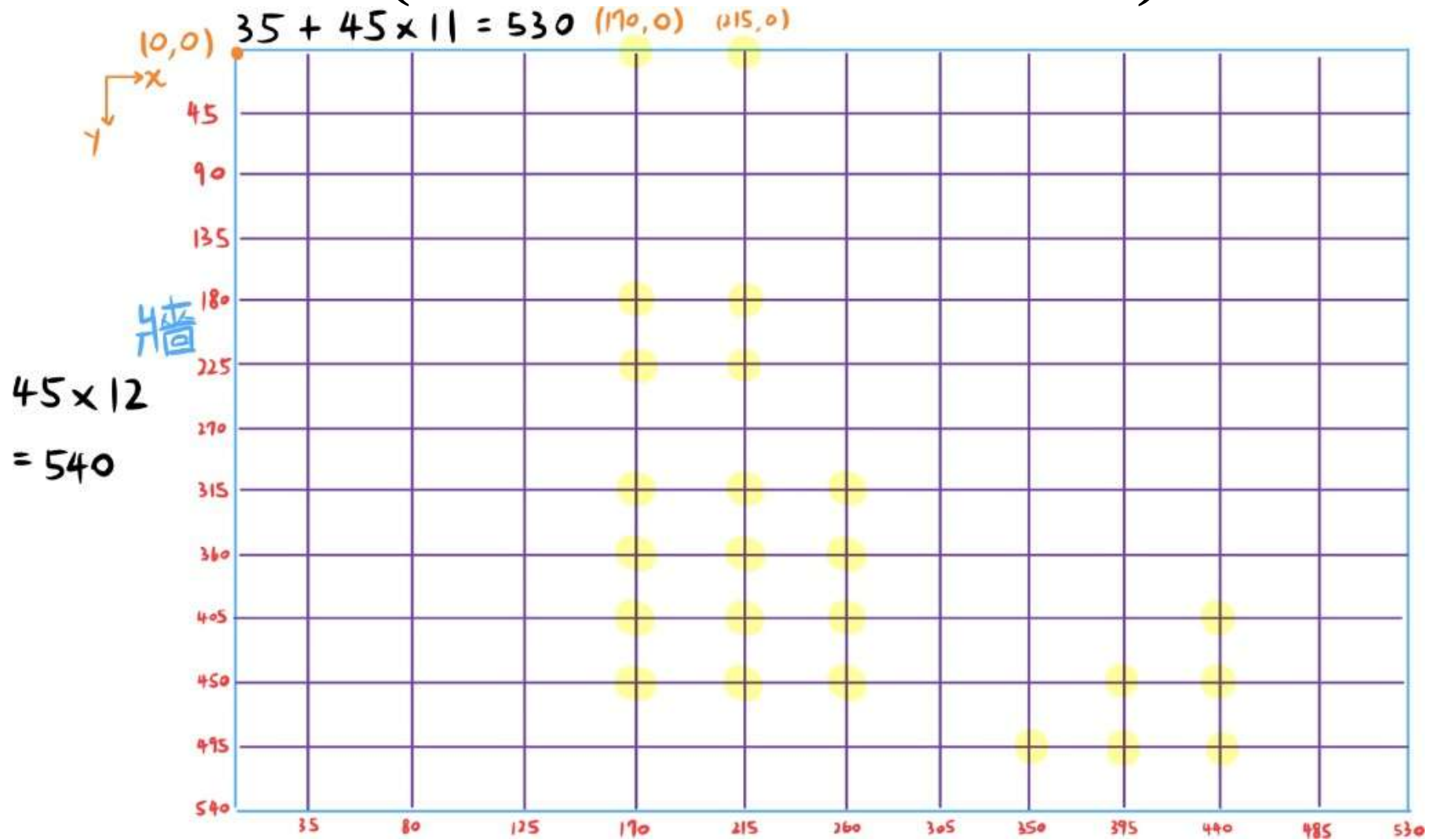


Spatial Alignment-(Target-based)

2026-07-23 16:31:38



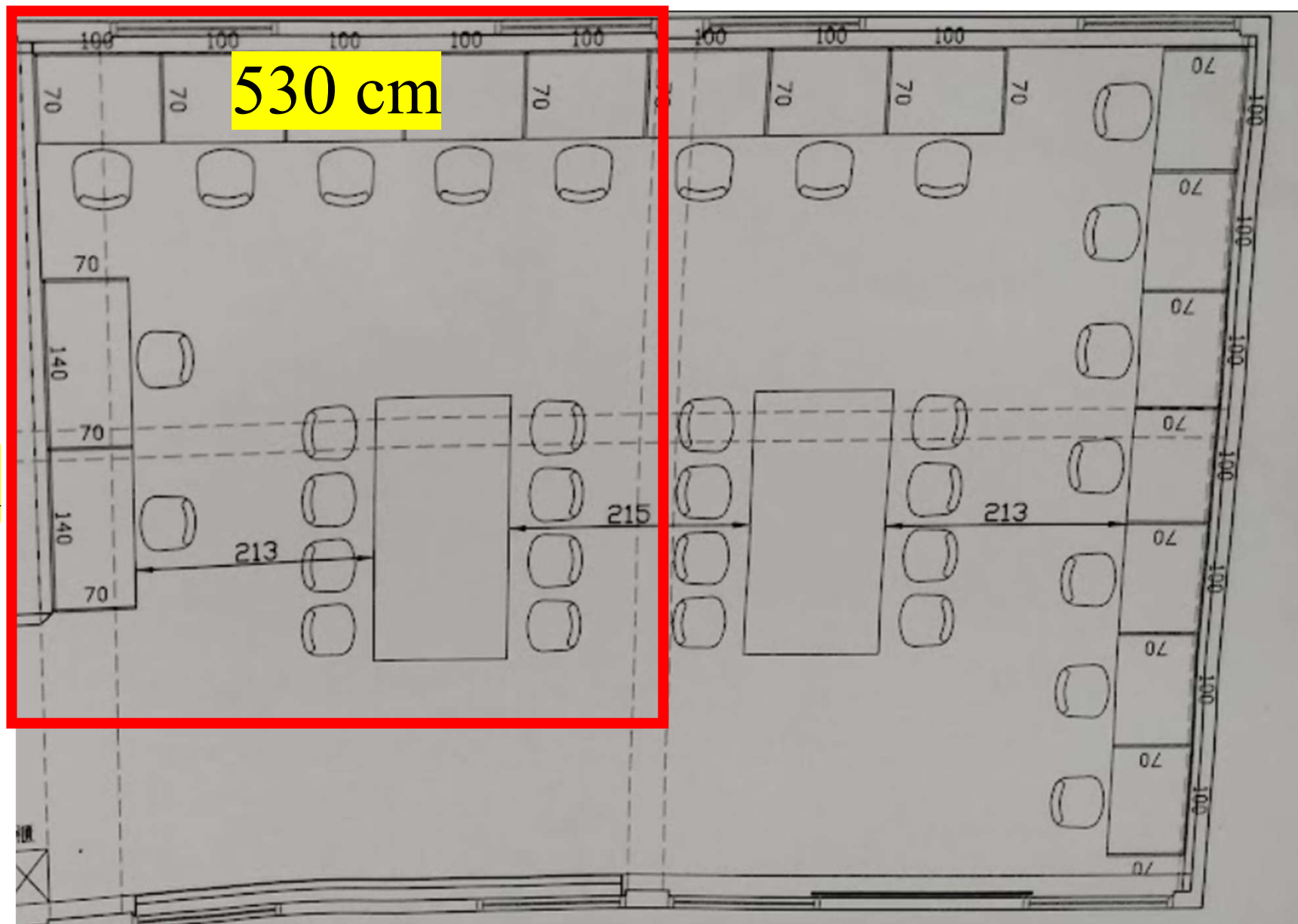
Calibration-(Define Coordinates)



平面圖

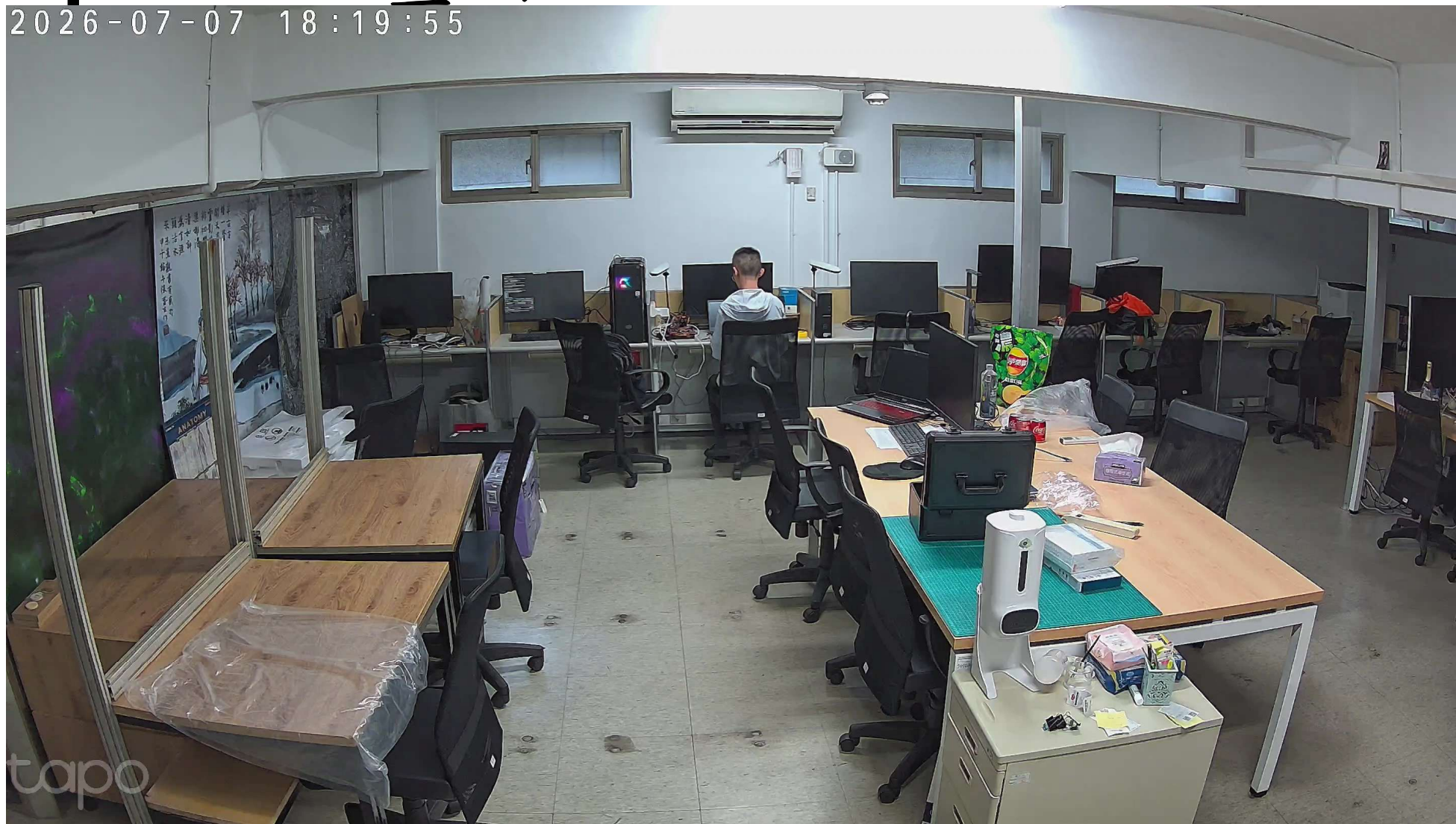
530 cm

540 cm



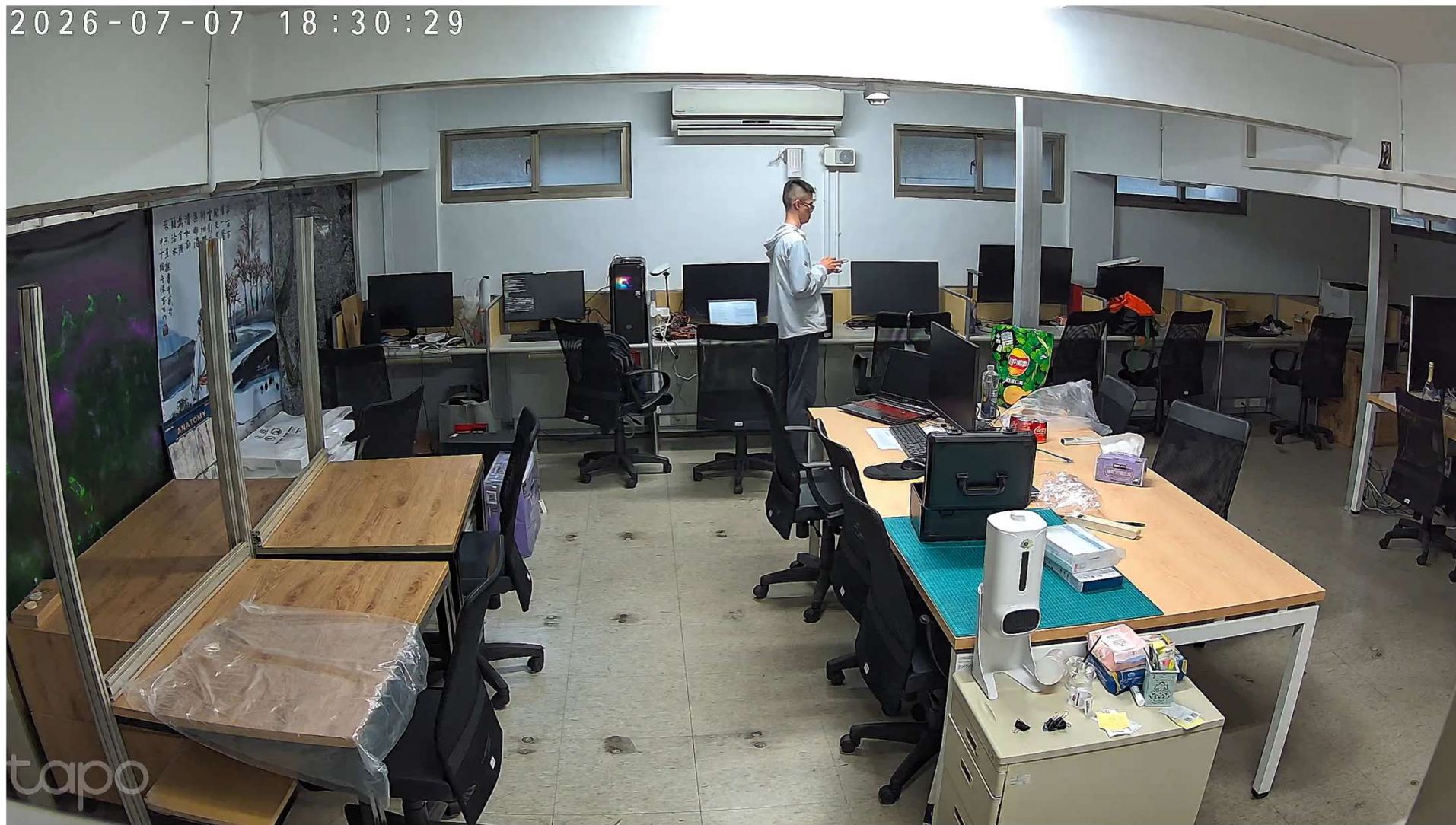
Tapo C230 畫面

2026-07-07 18:19:55

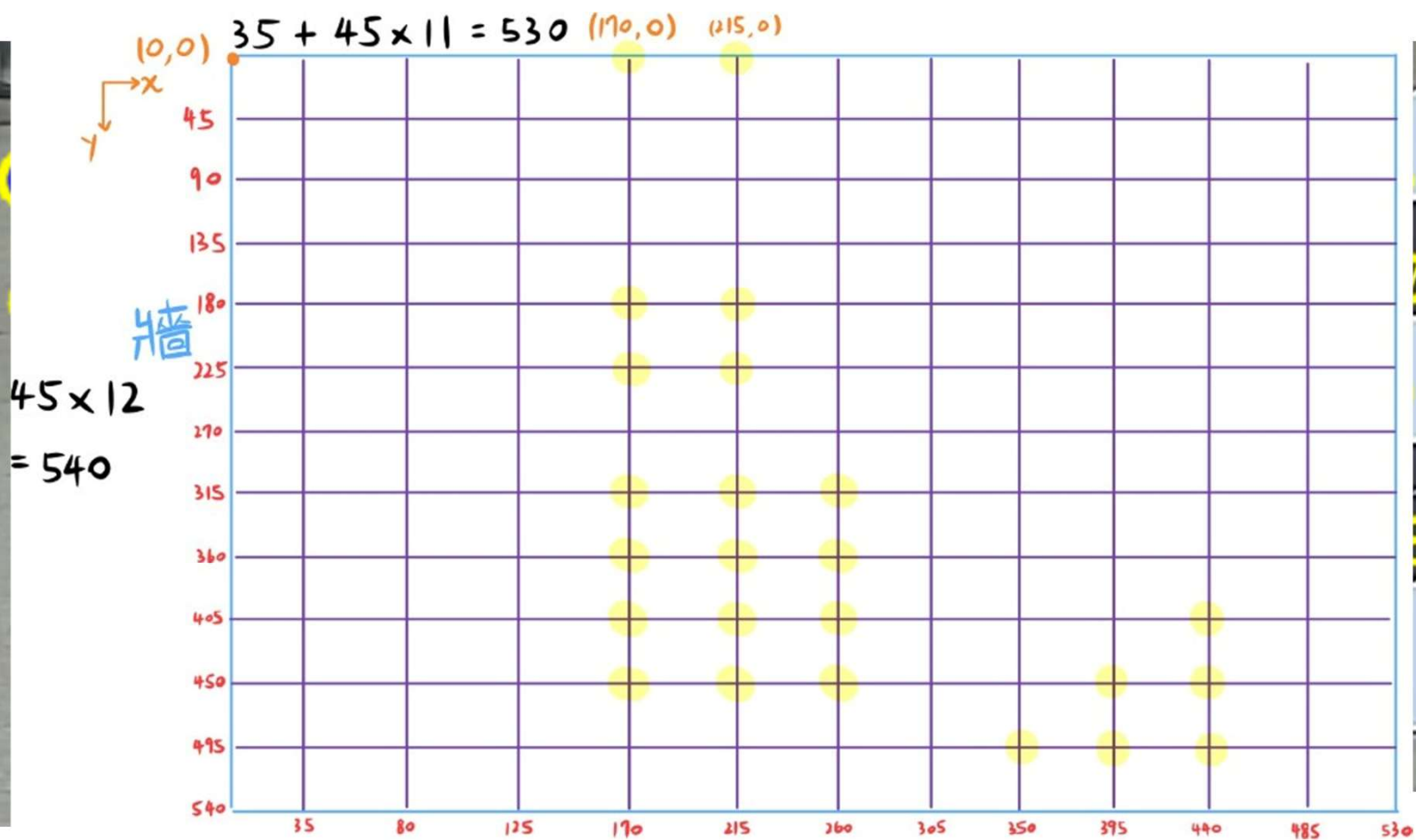


Tapo C230 即時畫面(適合用來測試)

2026-07-07 18:30:29



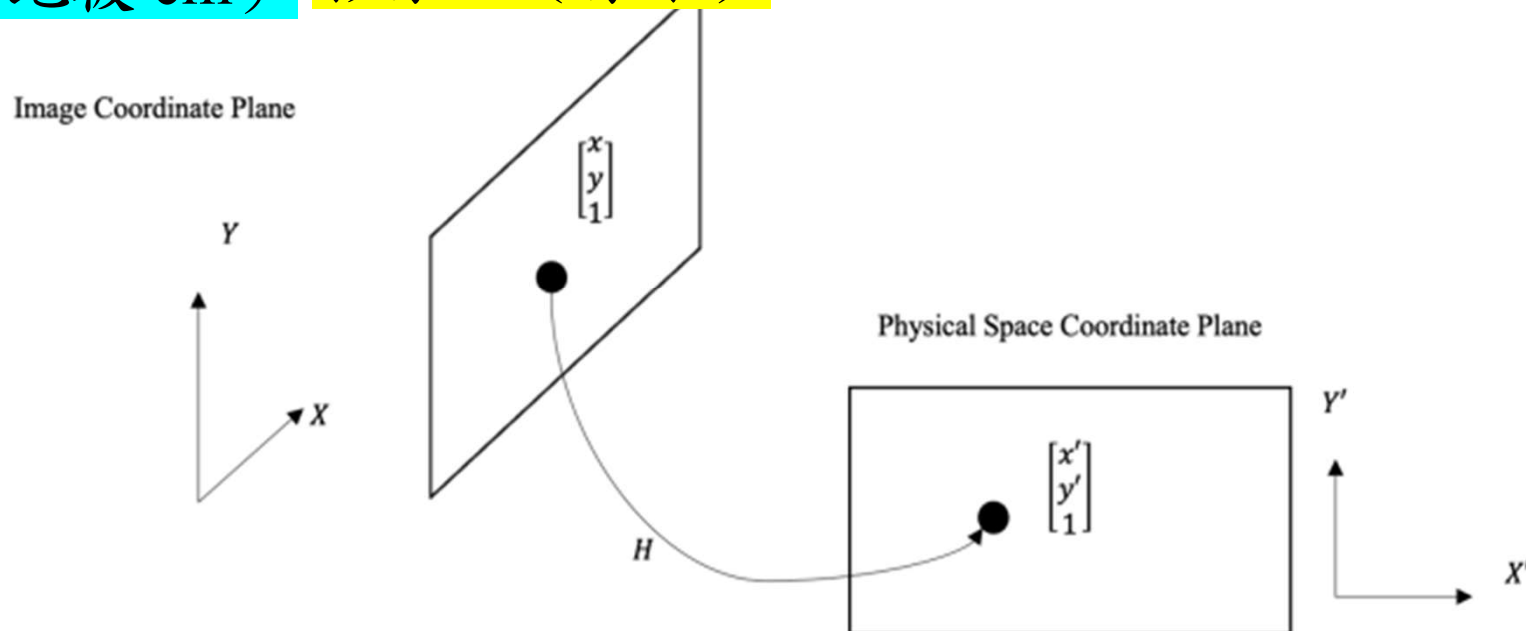
Calibration-(Set Anchor)



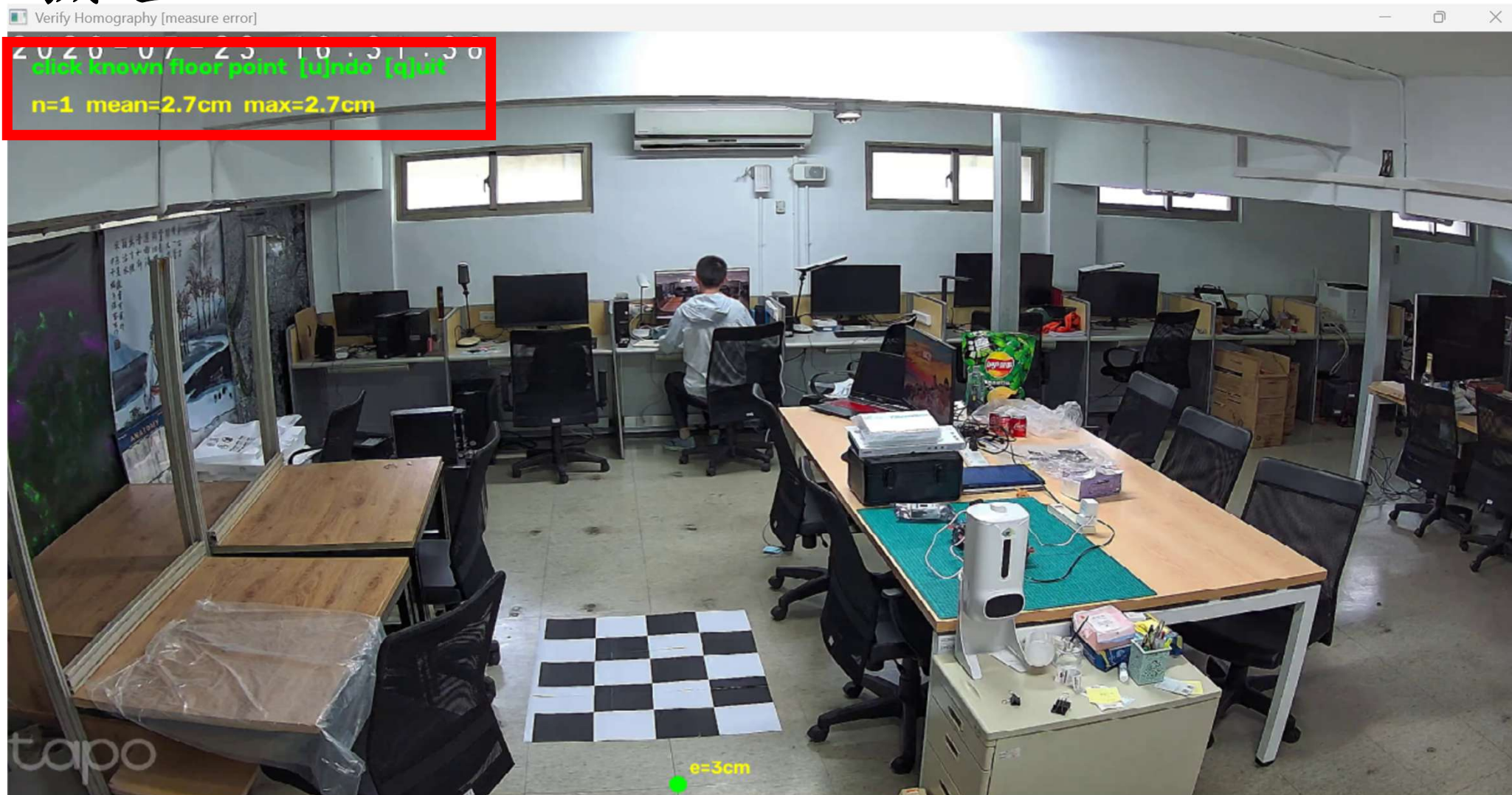
Calibration-(Homography Matrix)

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = H \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

世界座標 (地板 cm) 影像點 (像素)



驗證



Frame Capture

1. RTSP 降延遲:

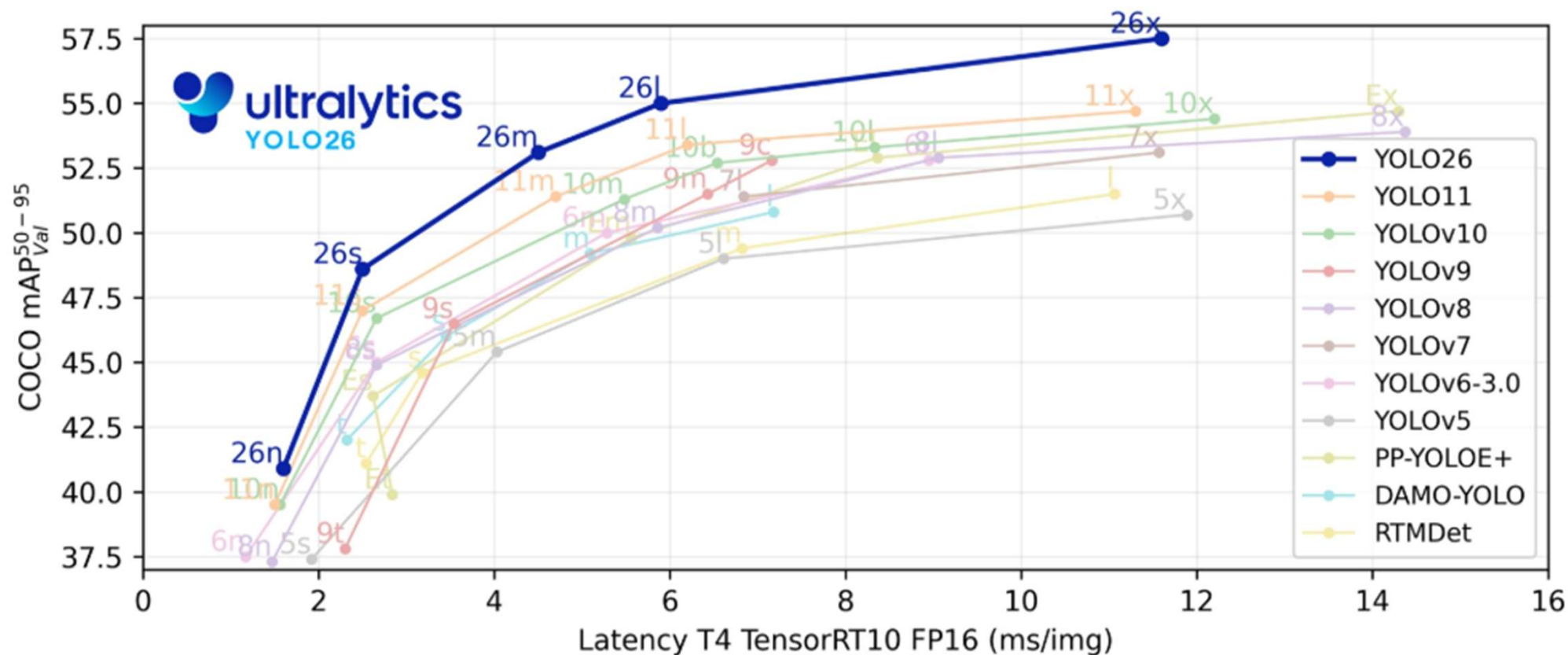
- 只處理最新幀 (LatestFrameCapture)
- RTSP 走 TCP

2. 跳幀 (--stride) :

RTSP／影片不必每幀都跑 YOLO，可跳幀降低運算量，中間幀沿用上次偵測結果：

Detection(YOLO 26s)

第一次測試時用YOLO v8

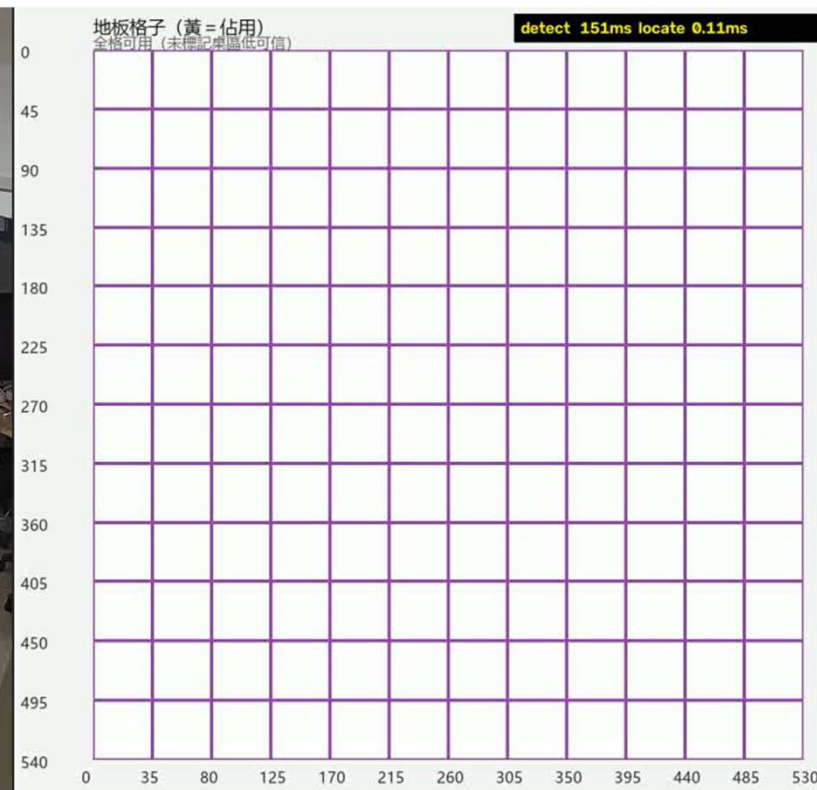
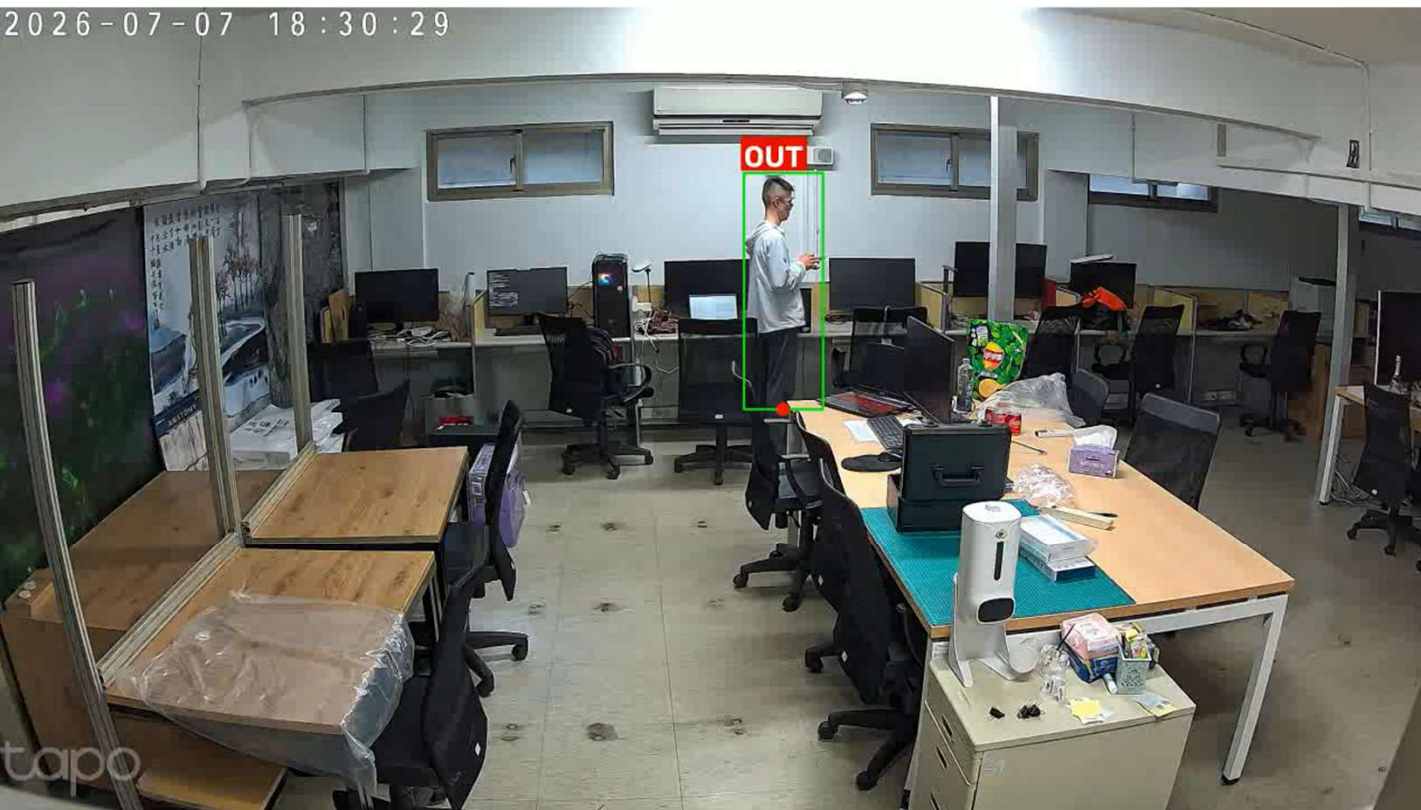


偵測 + 定位（腳點 → 格子）

模式	作法
foot	bbox 底邊中點（腳可見時）
head_drop	從頭往下，應付桌邊遮擋
auto（預設）	自動選 foot/head_drop

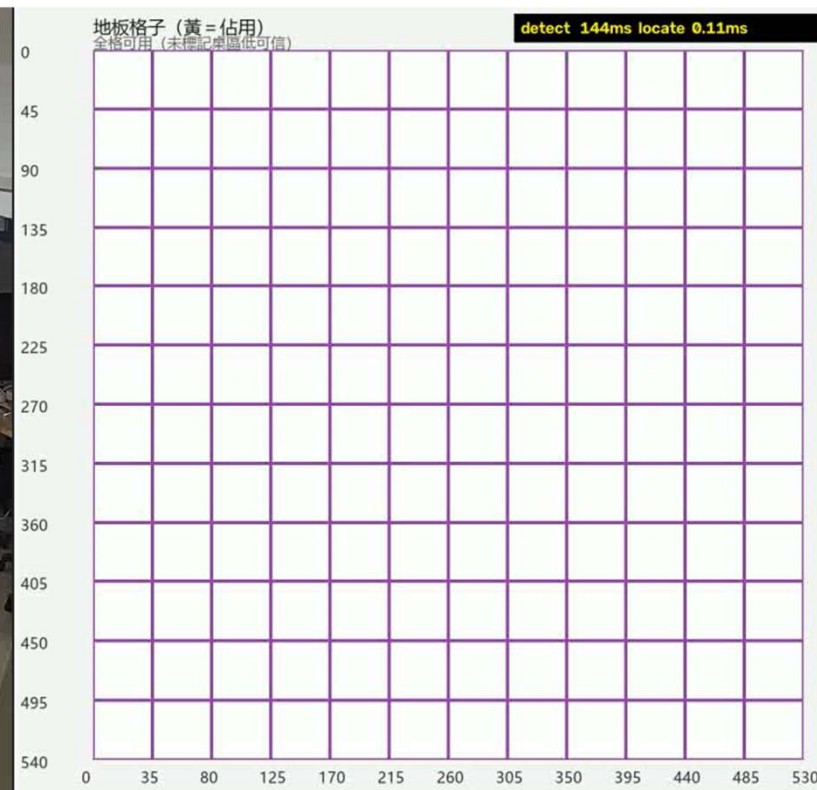
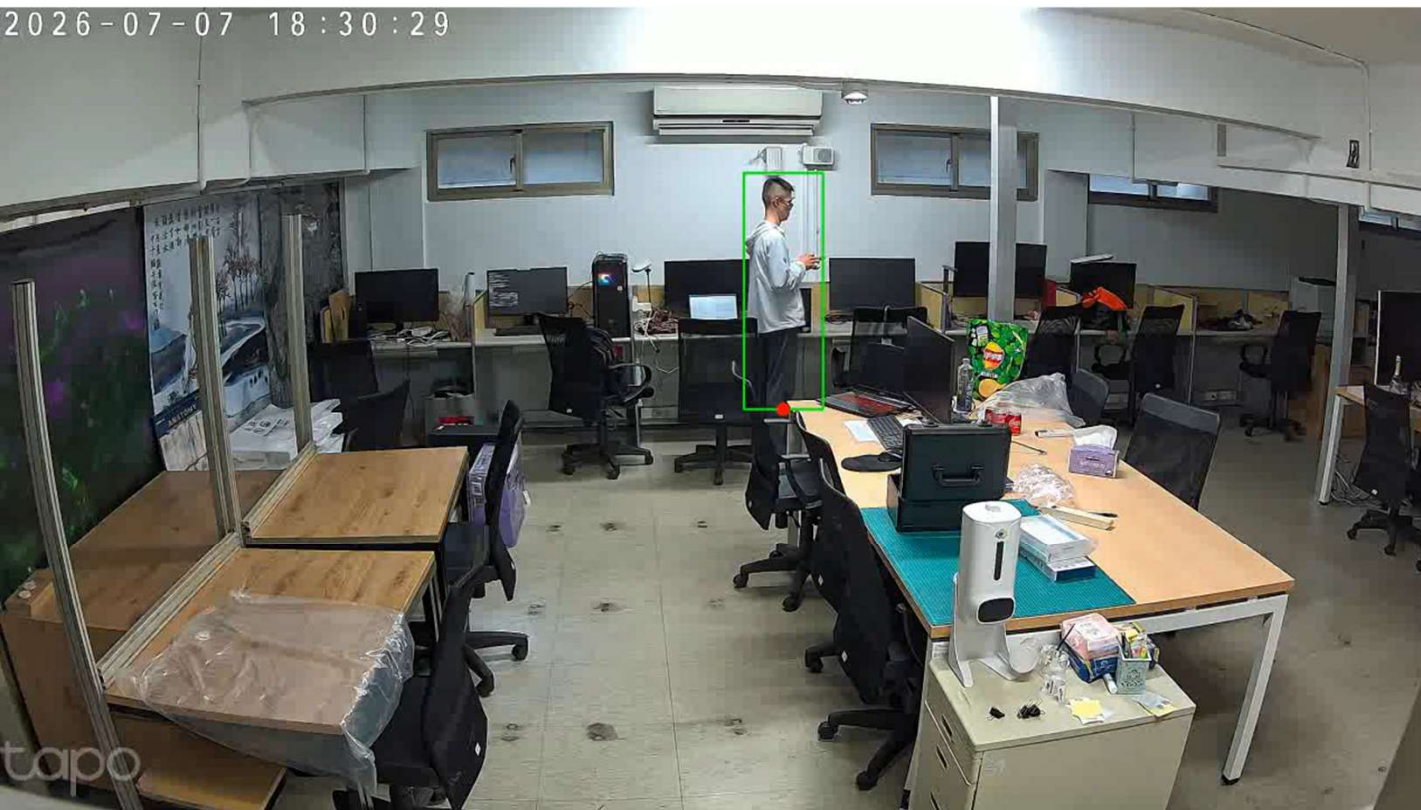
Result-(v1 , test.mp4)

2026-07-07 18:30:29



Result-(v2 , test.mp4)

2026-07-07 18:30:29



Real Time Test

時數紀錄

Week	Date	Hours	
Week 1	7/1 ~ 7/3	7	
Week 2	7/6 ~ 7/10	16	
Week 3	7/13 ~ 7/17	16	
Week 4	7/20 ~ 7/24	16	
Week 5	7/27 ~ 7/31	16	71
Week 6	8/3 ~ 8/7	16	
Week 7	8/10 ~ 8/14	16	
Week 8	8/17 ~ 8/21	16	48
Week 9	9/7 ~ 9/11	16	
Week 10	9/14 ~ 9/18	16	
Week 11	9/21 ~ 9/25	16	
Week 12	9/28 ~ 9/30	16	64
		Total :	183

感謝您的聆聽